

DM583 Data Mining – Example Questions for MCQ Exam

In both the **exercise sheets** as well as in **some of the lecture slides**, students can find a **reasonable sample of exercises** that they can use to help prepare for the final written exam. Below, three examples are provided to illustrate how the questions can be posed in a multiple-choice format where there are several answers presented, and some subset of these answers comprises the correct (true) answers:

Question 1 [XXX Points]

In the context of EM-GMM clustering, which of the following statements are correct?

1. EM-GMM clustering makes use of Bayes' rule/theorem.
☐ Yes/True ☐ No/False
2. EM-GMM clustering assumes independence between variables within clusters, so any pair of variables is theoretically expected to have zero covariance within a cluster.
☐ Yes/True ☐ No/False
3. Given an EM-GMM model with $k=2$ components (clusters), C_1 and C_2 , with probability densities as evaluated at a given point x equal to 0.0086 and 0.0136, respectively: if their prior probabilities are $\pi_{C_1} = 0.3$ and $\pi_{C_2} = 0.7$, then their posterior probabilities conditioned to point x are approximately 0.213 and 0.787, respectively.
☐ Yes/True ☐ No/False
4. Knowing that the covariance matrices of the two components are both diagonal, one can ascertain that the induced clusters will both necessarily be spherical, not elliptical.
☐ Yes/True ☐ No/False
5. One of the main advantages of EM-GMM over k-Means is that, unlike k-Means, EM-GMM is not sensitive to initialization of cluster centers.
☐ Yes/True ☐ No/False

Question 2 [XXX Points]

Consider a unidimensional dataset given by $\{-20, -10, 10, 40, 55, 95, 100\}$. Which of the following statements are correct?

1. If we want to perform *parametric* density estimation under the assumption that the data can be modelled by a single, univariate Normal distribution, application of the Maximum Likelihood Estimator (MLE) on the above dataset would result in a Gaussian pdf with optimal parameters given by the sample mean and sample variance, i.e., $\mu = (-20 - 10 + 10 + 40 + 55 + 95 + 100) / 7$, and $\sigma^2 = ((-20 - \mu)^2 + (-10 - \mu)^2 + (10 - \mu)^2 + (40 - \mu)^2 + (55 - \mu)^2 + (95 - \mu)^2 + (100 - \mu)^2) / 7$.
☐ Yes/True ☐ No/False
2. Without doing the computations, we can be certain that the results of hierarchical clustering of this dataset using Single-Linkage or Complete-Linkage will necessarily be the same because the dataset is unidimensional.
☐ Yes/True ☐ No/False
3. It is not possible to run the parallel/distributed version of k-means on this dataset because we have an odd number of observations (7 data points), so it is impossible to distribute the data evenly across the processing nodes, which is a requirement of the algorithm.
☐ Yes/True ☐ No/False
4. The ordinary (unweighted) version of the KNN outlier detection algorithm with hyper-parameter $k = 1$ would, for any choice of Minkowski distance, produce the following outlier scores for the seven data points: $\{10, 10, 20, 15, 15, 5, 5\}$.
☐ Yes/True ☐ No/False
5. With hyper-parameters $\epsilon = 15$ and $\text{MinPts} = 1$ or 2 , DBSCAN would in both cases produce a partition of the data with 4 clusters as follows: $C1 = \{-20, -10\}$, $C2 = \{10\}$, $C3 = \{40, 55\}$, $C4 = \{95, 100\}$.
☐ Yes/True ☐ No/False

Question 3 [XXX Points]

Consider the following database of transactions:

Transaction ID	ITEMS
1	BEER, CRACKERS
2	BREAD, BUTTER, DIAPERS, MILK
3	BEER, DIAPERS
4	BREAD, BUTTER, DIAPERS, MILK
5	BREAD, BUTTER, DIAPERS, MILK
6	BEER, CRACKERS, DIAPERS
7	BREAD, BUTTER, CRACKERS
8	BREAD, CRACKERS, DIAPERS, MILK
9	BREAD, BUTTER, MILK
10	BEER, BREAD, DIAPERS, MILK
11	BUTTER, DIAPERS
12	BREAD, BUTTER, DIAPERS

In the following, consider the Apriori algorithm with a minimum threshold for the support count equal to 5. Note that the support counts of the individual items present in the transactions are: BEER (4), BREAD (8), BUTTER (7), CRACKERS (4), DIAPERS (9), MILK (6). Further, note that, by setting the minimum support as 5, the *candidate* 2-itemsets obtained from the frequent 1-itemsets are:

<i>CANDIDATE 2-ITEMSETS</i>	SUPPORT
{BREAD, BUTTER}	6
{BREAD, DIAPERS}	6
{BREAD, MILK}	6
{BUTTER, DIAPERS}	5
{BUTTER, MILK}	4
{DIAPERS, MILK}	5

1. The Apriori algorithm will preliminarily produce the candidate 3-itemset {BUTTER, DIAPERS, MILK} from the candidate 2-itemsets above and then compute its support as 3 to determine that this is not a frequent itemset.
() Yes/True () No/False
2. The Apriori algorithm will preliminarily produce the candidate 3-itemset {BREAD, BUTTER, MILK} from the frequent 2-itemsets and then, the algorithm will have to

compute this candidate's support from the database before it can determine whether the candidate is a frequent 3-itemset or not.

☐ Yes/True ☐ No/False

3. Without looking at the database and only based on the above table of candidate 2-itemsets, we can ascertain that {DIAPERS, MILK} is a maximal frequent itemset.

☐ Yes/True ☐ No/False

4. {BREAD, DIAPERS} is a closed frequent itemset.

☐ Yes/True ☐ No/False

5. One of the possible association rules that follow from the frequent 3-itemsets in this database is $\{\text{DIAPERS}\} \Rightarrow \{\text{BREAD, MILK}\}$, with support equal to 5 and confidence equal to 5/9.

☐ Yes/True ☐ No/False